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dea-tools: Geospatial analysis of satellite data using Xarray, the Open Data Cube, and Digital Earth Australia

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Summary

`dea-tools` is an open-source Python library that provides a comprehensive set of tools for analysing, visualising, and modelling geospatial and Earth observation data represented as **xarray** objects. Originally developed to support **Digital Earth Australia (DEA)** workflows, `dea-tools` now functions as a general-purpose library that can be used with any geospatial dataset. The package offers high-level utilities for spatial and temporal analysis, remote sensing index calculation, machine learning, and interactive visualisation, while integrating seamlessly with the Python geospatial stack (**xarray**, **geopandas**, **numpy** and **dask**). Distributed as a **pip-installable environment**, `dea-tools` includes **JupyterLab** and related dependencies, providing a fully managed platform for scientific geospatial analysis on any machine. Embedded within the broader **`dea-notebooks`** repository, the package is supported by extensive documentation and real-world examples Jupyter notebooks demonstrating use of dea-tools functionality, along with longer analysis examples.Together, `dea-tools` and `dea-notebooks` enable reproducible, scalable, and accessible Earth observation science in Python.

Statement of need

Modern Earth observation workflows often involve complex, multi-temporal, and multi-resolution datasets that demand integrated tools for efficient processing and visualisation. While foundational libraries such as **xarray**, **geopandas**, and **dask** provide the building blocks, implementing complex analysis workflows still requires substantial coding effort to handle data loading and harmonisation, visualisation, analysis, and modelling. The `dea-tools` library addresses this gap by providing a curated suite of high-level, well-documented functions that seamlessly integrate with these core libraries.

The **`dea-tools`** Python package provides a comprehensive, open-source toolkit for the analysis, visualisation, and interpretation of geospatial and Earth observation (EO) data represented primarily as **xarray** objects. Developed by **Digital Earth Australia (DEA)** to streamline analysis of national-scale satellite datasets, `dea-tools` has since evolved into a flexible and general-purpose library for any geospatial workflow. It can now be used on any system and with any geospatial dataset, whether accessed from cloud-based catalogues via **ODC-STAC** or stored locally.

A major advantage of `dea-tools` is it comes bundled with all essential dependencies for scientific geospatial analysis, including **JupyterLab**, **xarray**, **geopandas**, **numpy**, **dask**, and other related ecosystem libraries. When deployed within a virtual environment, `dea-tools` therefore provides a **fully managed and ready-to-use geospatial analysis platform**. This design lowers the entry barrier for new users, enabling a self-contained environment for exploration, visualisation, and large-scale satellite data processing.

The package enables users to:

- Load, combine, and manage satellite or model datasets, including but not limited to DEA products, while handling reprojection and resolution harmonisation;
- Conduct **spatial and temporal analyses** directly on xarray objects (e.g., extracting temporal statistics, linear regression, and interpolation);
- Apply a large curated list of **remote sensing indices** such as NDVI, NDWI etc.
- Perform **machine learning and segmentation workflows** on xarray datasets, with built-in training, fitting, and prediction tools that work seamlessly with xarray;
- Generate **publication-quality visualisations and animations**, including RGB true and false colour composites and customisable time-lapse animations;
- Seamlessly convert between raster and vector formats using xarray–geopandas interoperation utilities.

The **broader `dea-notebooks` repository**, which includes `dea-tools`, provides extensive documentation as Jupyter notebooks including how-to guides and longer 'real-world' application examples. These include end-to-end workflows that demonstrate how to acquire satellite data via `odc-stac`, process it using `dea-tools`, and visualize outputs interactively within JupyterLab. Together, `dea-tools` and `dea-notebooks` offer a **reproducible, and fully managed framework** for geospatial analysis in Python — bridging the gap between foundational geospatial libraries and applied EO research.

Features

Below we highlight a non-exhaustive list of funtionality with dea-tools. Note that each of the functions listed below has an accompanying Jupyter notebook outlining how to utilise the function.

Analytical tools native to Xarray

Flexible and powerful functionality for scientific analysis with xarray objects:

- Extracting time series statistics, using the module dea-tools.temporal:
  - Generic summary statistics on any time series with `temporal_statistics`
  - Phenology with `xr_phenology`
  - Linear regression with `xr_regression`
- Validation tools within dea-tools.validation:
  - Compute common statistical metrics with `eval_metric`
  - Generate random points over an xarray object with `xr_random_sampling`

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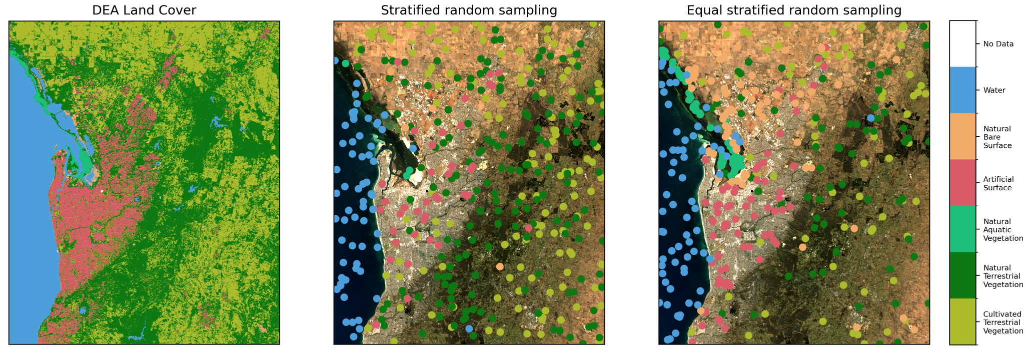

Figure 1: Example extraction of random samples from a classified xarray.DataArray using two of the sampling strategies available in the `xr_random_sampling` function.

- Spatial analysis tools within [dea-tools.spatial](#):
  - Vectorise xarray.DataArrays into geopandas.GeoDataFrames with `xr_vectorise`
  - Rasterize geopandas.GeoDataFrames into xarray.DataArrays with `xr_rasterize`
  - Extract subpixel contours from xarray.DataArrays with `subpixel_contours`
  - Interpolate point data stored in a geopandas.GeoDataFrame into an xarray.DataArray with `xr_interpolate`
- Xarray wrappers for the full machine learning pipeline in [dea-tools.classification](#):
  - prepare data for ML predictions with `sklearn_flatten` and `sklearn_unflatten`
  - Fit models with `fit_xr`
  - ML predictions with `predict_xr`
  - Handle spatial autocorrelation in cross-validation workflows with `spatial_train_test_split` and `SKCV` (spatial k-fold cross validation)

Data handling

Tools for loading and manipulating both DEA satellite data, along with other providers of geospatial data, stored within [dea-tools.datahandling](#)

- load_ard
- load_reproject
- parallel_apply

Plotting

The [dea-tools.plotting](#) library has extensive functionality for generating beautiful imagery from xarray objects.

- Highly customisable animations with `xr_animation`
- RGB plots for true and false colour composite images with `rgb`

Research projects

`dea-tools` and the broader `dea-notebooks` repository supports a wide range of scientific applications, from **agriculture and land cover mapping** to **wetland, coastal, and surface water monitoring**, and **climate and fire analysis**. The package has already underpinned more than **25 peer-reviewed studies**, demonstrating its robustness and adaptability across domains. Usage of the library is recorded [here](#).

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Thanks

References